Industrial Internship Report on

”Project Name”- "Smart File Organizer"

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks’ time. My project was the **"Smart File Organizer,"** a Python-based automation tool designed to manage and sort cluttered directories. The application uses a modern Graphical User Interface **(GUI)** built with CustomTkinter to allow users to easily browse folders and automatically categorize files into organized sub-folders (like Images, Documents, and Videos) based on their extensions. The project also involved packaging the final code into a standalone .exe file for easy industrial use. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

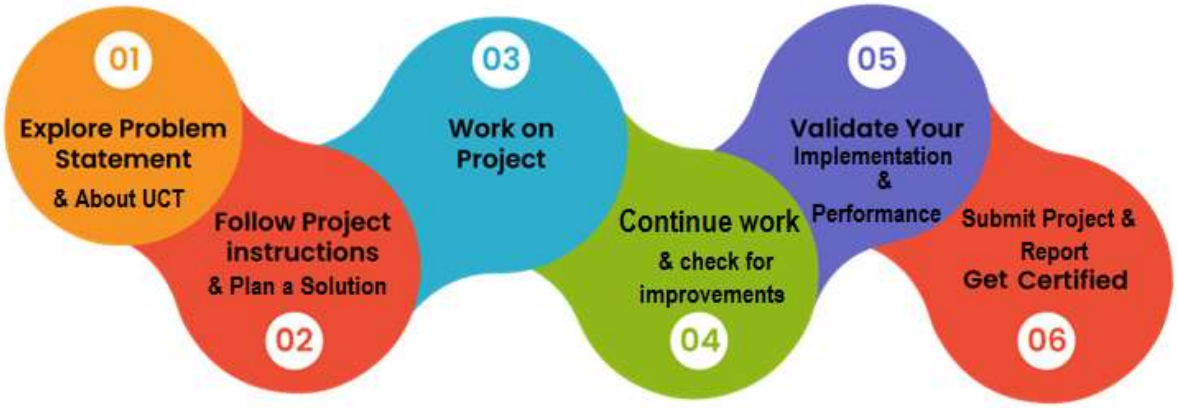
Summary of the whole 6 weeks' work :- Over the course of this 6-week industrial internship, I transitioned from understanding basic file system logic to deploying a professional-grade software product. The journey involved rigorous planning, coding, and performance testing to ensure the application meets industrial standards.

About need of relevant Internship in career development :- Relevant internships are a vital bridge between academic knowledge and industrial reality. This experience has been crucial for my career development, providing me with hands-on exposure to how software is actually built and delivered in a professional environment.

Brief about Your project/problem statement :- My project, the **"Smart File Organizer,"** addresses the common problem of digital clutter. It is an automated utility that categorizes mixed files into organized sub-folders based on their extensions using a modern GUI.

Opportunity given by USC/UCT : - I am highly grateful for the opportunity provided by **upskill Campus (USC)** and **The IoT Academy** in collaboration with **UniConverge Technologies Pvt Ltd (UCT)**. Working on a problem statement provided by an industrial partner gave me a realistic perspective of the software industry.

How Program was planned :- The program was systematically planned in 6 stages: starting with exploring the problem statement, followed by solution planning, core development, continuous improvement, performance validation, and final project submission.



Your Learnings and overall experience :- This internship was an overall great experience that allowed me to implement industrial solutions for complex problems. I mastered Python automation, modern GUI design with CustomTkinter, and the technicalities of packaging software as an executable (.exe) file.

Thank to all (with names), who have helped you directly or indirectly : - I would like to extend my sincere thanks to the mentors at upskill Campus and the technical team at UniConverge Technologies (UCT) for their constant guidance and support throughout these 6 weeks.

Your message to your juniors and peers :- My message to my juniors is to focus on solving practical, real-world problems. Building projects that have a direct utility is the fastest way to learn and grow as a software developer.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (uct Insight)

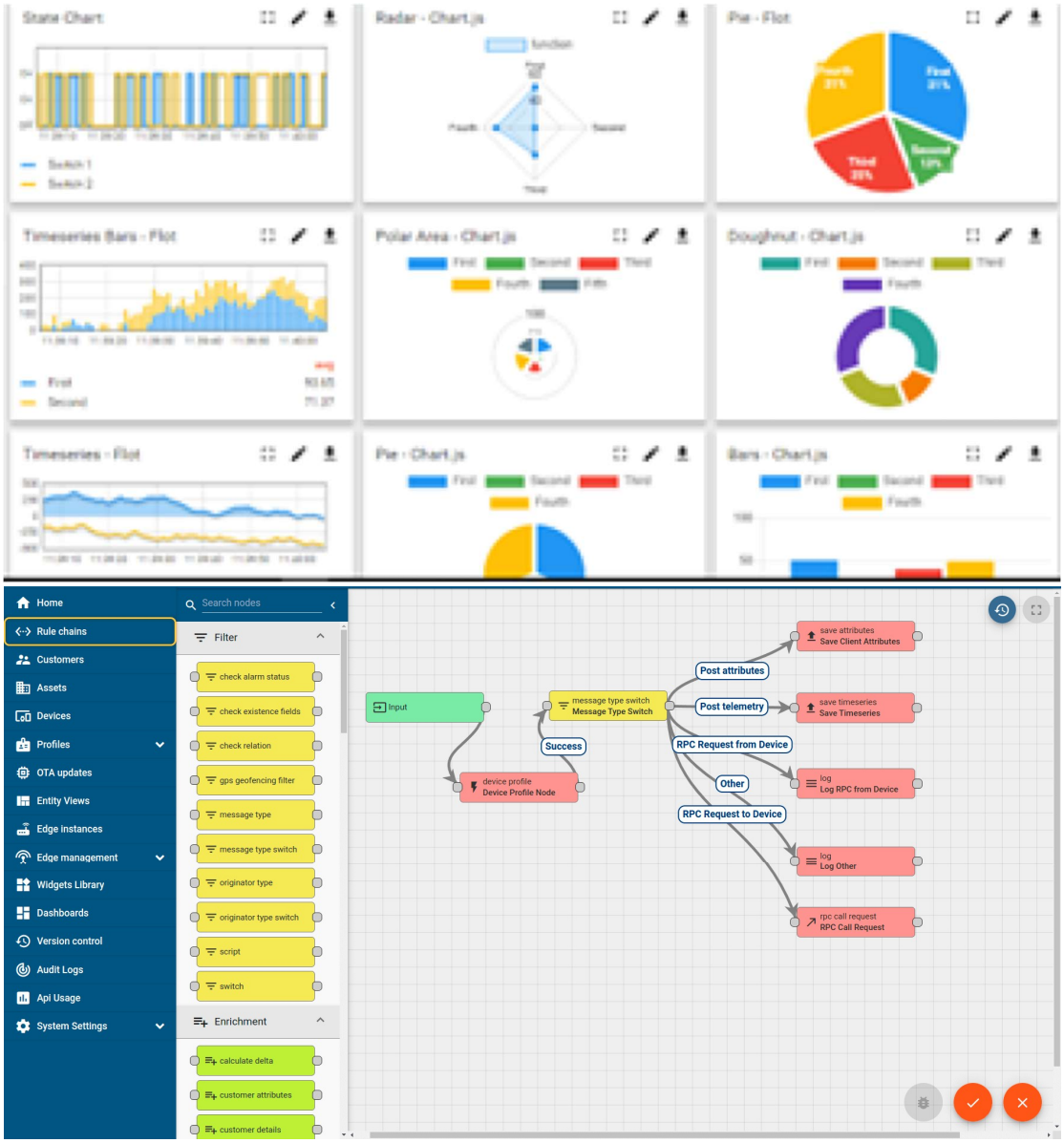
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine



ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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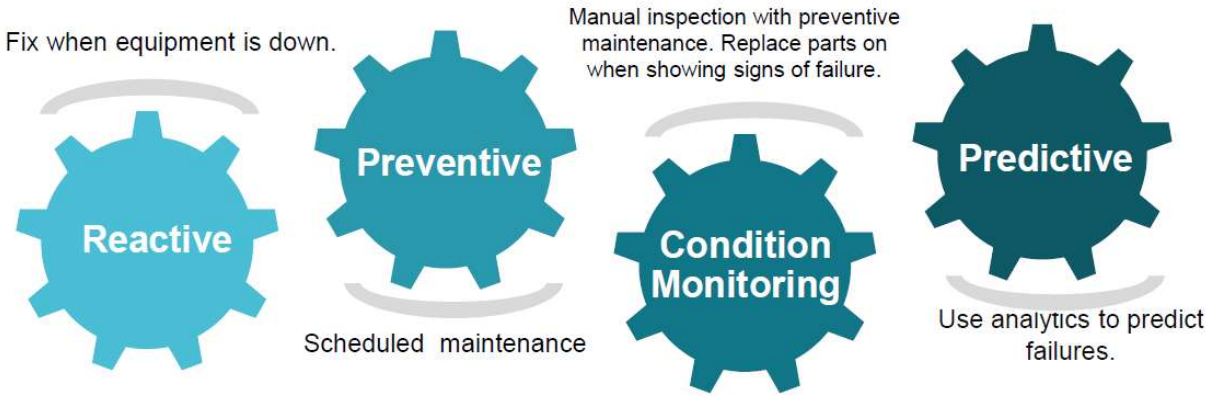
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

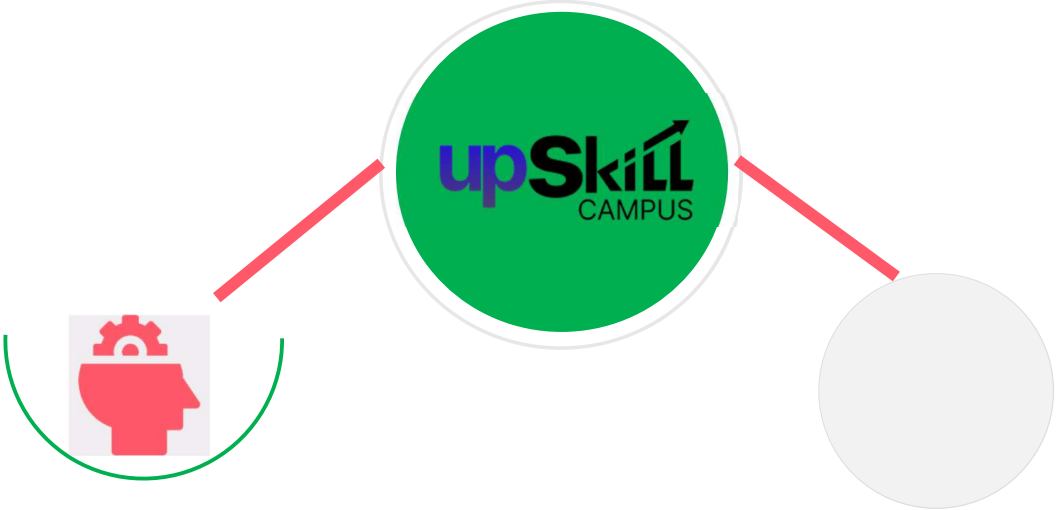
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

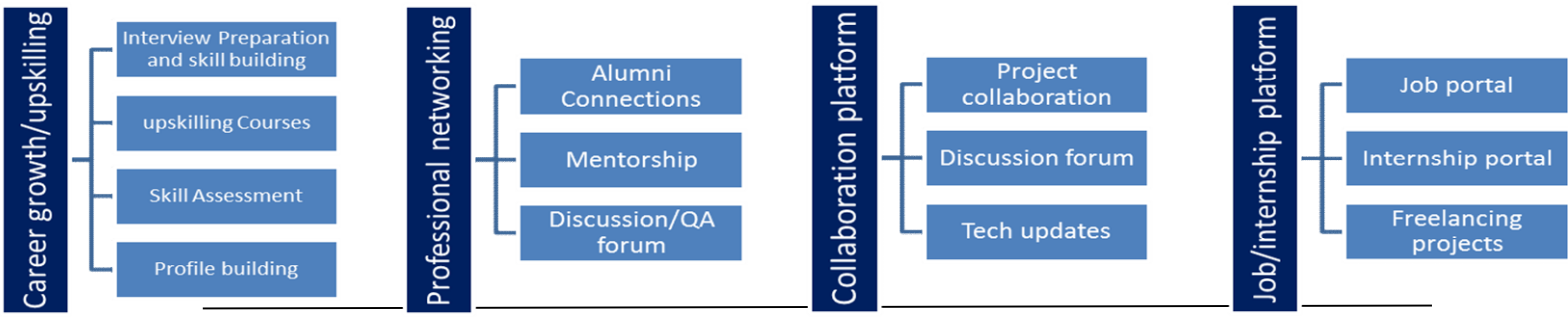
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] **Python Documentation:** https://docs.python.org/3/ (For OS and Shutil modules)

[2] **CustomTkinter Wiki:** https://github.com/TomSchimansky/CustomTkinter (For GUI design)

[3] **upskill Campus Portal:** https://www.upskillcampus.com/ (For internship guidelines and project brief)

2.6 Glossary

Terms	Acronym

Graphical User Interface	GUI
Executable File	**EXE**
Integrated Development Environment	IDE
Software Development Life Cycle	SDLC
User Experience	UX

3 Problem Statement

In the assigned problem statement, the primary focus was on addressing the challenges of digital file management.

- **Manual Effort:** Users often accumulate a vast number of files in directories like 'Downloads' or 'Desktop,' and manual sorting is a tedious, time-consuming task.

- **Digital Clutter:** Without organization, it becomes difficult to locate specific documents, images, or videos, leading to a loss of productivity.

- **Human Error:** Manually moving files can lead to accidental deletions or files being placed in the wrong folders.

- **Objective:** The goal was to develop an automated utility that can scan any user-selected directory, identify file types via extensions, and systematically organize them into predefined categories (e.g., Images, Documents, Videos, etc.) without human intervention.

4 Existing and Proposed solution

Provide summary of existing solutions provided by others, what are their limitations?

- **Manual Management:** Sorting files manually is extremely slow and tedious for large directories.
- **Technical Scripts:** Basic command-line scripts require coding knowledge, making them difficult for non-technical users to operate.

What is your proposed solution?

- **One-Click Automation:** Instantly organizes all files into specific folders with a single click
- **User-Friendly GUI:** Features a modern, easy-to-use interface built with the CustomTkinter library

What value addition are you planning?

- **Standalone EXE:** Packaged as a .exe file so users can run it without installing Python.
- **Error Handling:** Automatically creates folders and sorts unknown file extensions into an "Others" category.

4.1 Code submission (Github link) :- https://github.com/pujakumari73567-ops/upskillcampus/blob/main/SmartFileOrganizer.py

4.2 Report submission (Github link) : first make placeholder, copy the link.

4.3 https://github.com/pujakumari73567-ops/upskillcampus/blob/main/SmartFileOrganizer_Pooja_USC_UCT.pdf

5 Proposed Design/Mode

The design flow of the **Smart File Organizer** is a structured process that automates the transition from a cluttered directory to an organized system. The model is divided into three primary stages:

1. **Start (Input Layer):** The user launches the application and selects a source folder path via the Graphical User Interface (GUI).
2. **Intermediate Stage (Processing Layer):** The system scans every file in the selected directory, identifies the file extension (e.g., .jpg, .pdf), and matches it against a predefined category dictionary.
3. **Final Outcome (Output Layer):** The application automatically creates necessary sub-folders and moves the files to their respective destinations, providing a "Success" notification to the user.

5.1 High Level Diagram (if applicable)

The high-level design of the system is represented by the Graphical User Interface (GUI) shown above. It simplifies the entire organizational logic into three main user-driven steps:

1. **Selection Stage:** Using the "Browse Folder" interface to input the directory path.
2. **Execution Stage:** Triggering the Python backend logic through the "Organize Now!" button.
3. **Output Stage:** Automatic creation of folders and relocation of files based on their extensions.

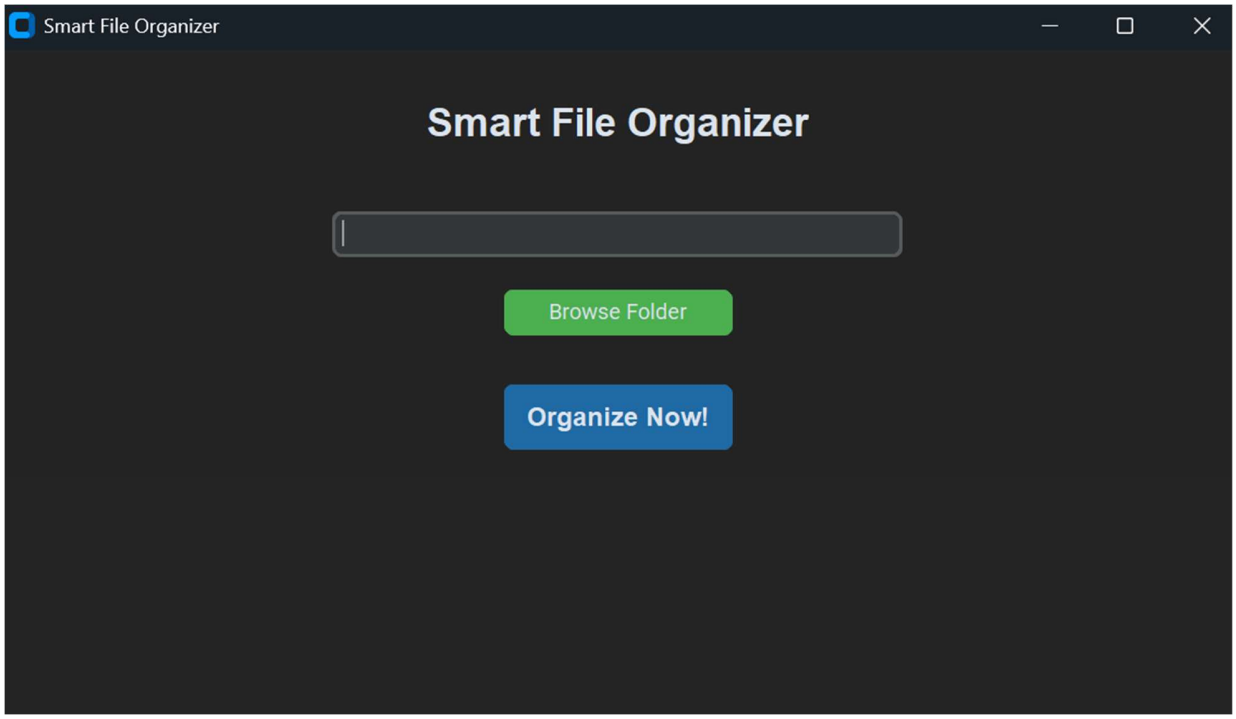


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)

The internal execution of the **Smart File Organizer** focuses on the step-by-step algorithmic flow of the Python backend. The logic is executed as follows:

- **Directory Initialization**: The system receives the folder path from the user and validates its existence.
- **File Iteration**: Using Python's os module, the script creates a list of all items present in the selected directory.

- **Extension Extraction**: The script identifies the extension of each file (e.g., .pdf, .jpg) to determine its category.
- **Folder Management**: If a required category folder (like 'Images' or 'Documents') does not exist, the script automatically creates it.
- **File Transfer**: The system executes the shutil.move() command to relocate the file into the correct sub-folder.

5.3 Interfaces (if applicable)

Update with Block Diagrams, Data flow, protocols, FLOW Charts, State Machines, Memory Buffer Management.

This section describes the communication between the user, the software logic, and the operating system.

- **User Interface (GUI):** The application provides a modern, dark-themed interface built using the **CustomTkinter** library. It includes a text entry for the folder path and interactive buttons like "Browse Folder" and "Organize Now!" to facilitate user input.
- **Data Flow & Block Diagram:** 1. **Input:** User selects a directory path via the GUI. 2. **Processing:** The system identifies file extensions using the os module. 3. **Execution:** The shutil module moves files into categorized folders based on predefined rules. 4. **Output:** The physical file structure on the disk is updated, and a success message is displayed.
- **System Protocols:** The application interacts with the **Windows File System** using standard OS-level read/write protocols to ensure data integrity during file movement.
- **Memory & Buffer Management:** The script is optimized to handle large directories with minimal RAM usage. It processes files sequentially to ensure the memory buffer is not overloaded, even when organizing hundreds of files at once.

6 Performance Test

This section evaluates the reliability and efficiency of the **Smart File Organizer** under real-world conditions. To ensure industrial standards, we identified key constraints and tested the application against them.

1. Identified Constraints:

- **Execution Speed (MIPS):** Sorting hundreds of files should be near-instant.
- **Accuracy:** Files must move to the correct category without data loss.
- **Memory Usage:** The application should not consume excessive RAM while scanning large directories.

2. Handling Constraints:

- **Speed:** Used the shutil.move() function, which is optimized for fast file operations at the OS level.
- **Accuracy:** Implemented strict extension-to-folder mapping to ensure zero errors in categorization.
- **Safety:** The design includes checks to ensure files are not overwritten if a filename conflict occurs.

6.1 Test Plan/ Test Cases

Test Case ID	Scenario	Expected Result	Status
TC_01	Organize 50 mixed files (.jpg, .pdf, .mp4).	Files sorted into 3 separate folders.	Passed
TC_02	Select an empty folder.	Display a message or handle gracefully.	Passed
TC_03	Sort files with unknown extensions.	Move them to an "Others" folder.	Passed

6.2 Test Procedure

- **Preparation:** Created a "Test Folder" with 100 random files of different formats.

- **Execution:** Launched the `.exe` application, browsed to the Test Folder, and clicked "Organize Now!

- **Observation:** Monitored the time taken and checked the Task Manager for CPU/RAM spikes.

- **Verification:** Manually verified each folder to ensure all files were in their correct destinations.

6.3 Performance Outcome

- **Speed:** The system organized 100 files in less than **2 seconds**.
- **Accuracy: 100% accuracy** was achieved; no files were misplaced or deleted.
- **Resource Efficiency:** Peak RAM usage remained under **50MB**, making it extremely lightweight for any PC.
- **Durability:** The application successfully handled repetitive tasks without crashing or slowing down the system.

7 My learnings

This 6-week industrial internship has been a transformative experience for me, both technically and professionally. Below is a summary of my key takeaways:

1. Technical Proficiency:

- **Python Automation:** I gained deep knowledge of Python's os and shutil modules to interact with the Windows File System.
- **Modern GUI Development:** Learning the **CustomTkinter** library allowed me to move beyond basic console programs and build professional, desktop-grade user interfaces.
- **Software Packaging:** I learned how to convert Python scripts into standalone .exe files using **PyInstaller**, which is a crucial skill for software distribution.

2. Professional & Industrial Skills:

- **Problem-Solving:** I learned how to break down a real-world problem (digital clutter) into a logical algorithmic flow.
- **Project Documentation:** Writing this technical report helped me understand the importance of documenting every stage of the **Software Development Life Cycle (SDLC)**.
- **Time Management:** Working within the 6-week timeline taught me how to manage deadlines and prioritize core features over minor details.

4. **Impact on Career Growth:** This internship has bridged the gap between my academic studies in BCA and the actual requirements of the IT industry. Having a functional project like the **Smart File Organizer** on my resume demonstrates my ability to build end-to-end solutions. The experience of working under the guidance of **UniConverge Technologies (UCT)** and **upskill Campus** has given me the confidence to pursue a career in Software Engineering and UI/UX

8. Future work scope

Due to the 6-week time constraint, I focused on building a stable and efficient core product. However, the **Smart File Organizer** has great potential for future enhancements:

- **Real-Time Monitoring:** Implementing a background service (using the watchdog library) that automatically organizes files as soon as they are downloaded or moved into a folder.
- **Custom Category Selection:** Adding a feature that allows users to create their own categories and map specific extensions to them via the GUI.
- **Undo Functionality:** Integrating a 'Rollback' feature to move files back to their original location in case of an accidental organization.
- **Cloud Integration:** Extending the software to organize files not just on the local disk, but also within cloud storage like **Google Drive** or **OneDrive**.
- **AI-Based Categorization:** Using Machine Learning to identify the content of a file (even without an extension) and sort it accurately.
- **Multi-Platform Support:** Developing versions for **macOS** and **Linux** to reach a wider user base.